

Lung Cancer

Joy Li, Francisco Chen, Hannah Higgins, Harper Lau

Introduction

Lung cancer is the uncontrollable cell growth (tumors) within the lung tissue. The most common causes of lung cancer are smoking, air pollutants, and genetics. A person with lung cancer may experience symptoms such as persistent coughing, chest pain, or shortness of breath. There are two main types of lung cancer: small cell lung cancer and non-small lung cancer. The type of lung cancer is determined by the tumors' cells shape and size. Lung cancer stages are classified into Stage 0, Stage I, Stage II, Stage III, and Stage IV, with each successive stage meaning the cancer is becoming more widespread throughout the body. Testing for lung cancer is done through X-rays and CT-scans, followed by diagnosis through biopsies. Treatments of lung cancer include surgery, chemotherapy, and radiation therapy. The best type of treatment is chosen based on the type of lung cancer and its stage. To prevent developing lung cancer, it is best to avoid smoking/vaping, reduce exposure to air pollution, and maintain an overall healthy lifestyle with good eating and exercise habits.

Causes

Lung cancer originally develops when a mutation occurs in a previously normal cell and leads to unregulated cell division, resulting in the production of too many cells. The abnormal multiplication of cells eventually forms a mass called a tumor. There are several causative factors of developing lung cancer. The most common cause of lung cancer, being responsible for about 80% of lung cancer deaths, is smoking. Cigarettes contain over 7,000 chemicals, many of which are known to cause cancer in people and animals and are considered toxic (Centers for Disease

Control and Prevention, n.d.). Scientists believe people who regularly smoke cigarettes are more likely to develop lung cancer than those who don't because cigarette smoke damages the cells that line the lungs. Although it is not known whether they are linked to lung cancer yet, smoking e-cigarettes and the usage of vapes have been shown to have damaging effects on lungs which could possibly lead to later health problems. Additionally, exposure to air pollutants such as radon and asbestos can increase risk of lung cancer. Radon is a naturally occurring radioactive gas which results from the breakdown of uranium in rocks and soil. When a high level of radon becomes concentrated in air spaces and is breathed into the body, it harms the lungs slightly. Over time, this small damage can lead to severe damaging of the lung cells' DNA, causing abnormal cell growth and tumors (MD Anderson Cancer Center, 2023). Asbestos is a group of naturally occurring fibrous minerals that are found in commercial products like insulation, fireproofing materials, and wallboard materials. If a product containing asbestos is disturbed, tiny asbestos fibers are released into the air and can become trapped in the lungs once breathed in. After several years the buildup of asbestos fibers can cause tissue inflammation and scarring, which can lead to serious health problems and an increased risk of lung cancer (National Cancer Institute, 2022). Another cause of lung cancer is genetics. People who have two or more first-degree relatives who have had lung cancer are at a higher risk due to hereditary genes.

Tests

Lung cancer happens when cells in the lungs grow in a way that is abnormal and out of control. Doctors divide lung cancer into two main types. One type is non-small cell lung cancer (NSCLC). This type is the most common. The second type is small-cell lung cancer (SCLC). This type is less common but often grows and spreads faster. The two types act differently in the

body. However, most patients move through a similar series of steps. A person notices symptoms, or a scan shows a “spot” in the lung. The doctor orders tests. The healthcare team makes a diagnosis and finds the stage. Then the team and the patient decide on treatment together.

Many countries now suggest low-dose CT (LDCT) screening for people with a high risk of lung cancer. This group often includes long-time smokers. LDCT is a special CT scan. It uses less radiation than a regular CT scan but can still find very small spots early. When a doctor worries about lung cancer, the doctor first asks about the person’s health history and does a physical exam. The doctor then orders imaging tests. A chest X-ray is quick and simple but can miss small tumors. A CT scan gives clearer and more detailed pictures of the lungs and nearby lymph nodes. Sometimes the team also uses a PET-CT scan. This test combines CT pictures with a PET scan that shows very active areas that may be cancer. If there is concern about spread to the brain or spinal cord, the doctor may order an MRI of the brain.

Scans give many clues, but they cannot prove that a lung mass is cancer. To be sure, doctors need a biopsy. A biopsy means that a small piece of tissue or a few cells is taken from the body and sent to a lab. A specialist called a pathologist looks at the sample under a microscope. The pathologist decides if the sample contains cancer. The pathologist also tells which type of lung cancer it is. There are several ways to get a biopsy sample. One common way is bronchoscopy. In bronchoscopy, the doctor passes a thin, flexible tube with a light and camera through the mouth or nose into the airways. If the doctor can see a tumor, the doctor can remove tiny pieces of it. For nodules near the outside of the lung, the team may use a needle biopsy. In a needle biopsy, the doctor uses CT images to guide a needle through the chest wall to the spot in the lung. Sometimes a surgeon needs to remove a small piece of lung through a cut in the chest.

This is a surgical biopsy. In a few cases, cells in mucus that a person coughs up can also show cancer. This test is called sputum cytology.

Stages

The staging system that is most often used for NSCLC (non-small cell lung cancer), a type of lung cancer, is the TNM system. There are three factors to it, and the doctors use them to describe how far NSCLC has spread.

“**T**” stands for “Tumor”, this category determines:

- How big is the tumor in the lung?
- Has it grown into nearby tissues or organs?

“**N**” stands for “Nodes” (Lymph Nodes), this category determines:

- Has the cancer spread to nearby lymph nodes?

“**M**” stands for “Metastasis”, this category determines:

- Has the cancer spread to other parts of the body, like the brain, bones, liver, or the other lung?

Each letter (T, N, and M) is followed by a number or letter that gives more details. Higher numbers mean the cancer is more advanced. After the doctor knows the T, N, and M values, they combine the information and assign an overall stage.

Non-small cell lung cancer (NSCLC) is divided into five stages. **Stage 0** is the earliest form. In this stage, there are usually no symptoms, because the cancer cells are limited to the lining of the airway and have not yet invaded deeper into the lung tissue. Stage 0 is mostly considered as a pre-invasive lesion and not counted as “real cancer”. **Stage I** is when the cancer

is small (no more than 3 cm) and limited to the lung. At this point, the tumor has not spread to any lymph nodes, which makes it easy to treat. **Stage II** is when the tumor has grown larger. In this stage, the cancer could spread to nearby nodes depending on the different possible stage groupings. Although the cancer is still cuddled together, treatment becomes more complex. **Stage III** is when the cancer has spread to the lymph nodes in the center of the chest. Doctors often divide this into Stage 3A and Stage 3B. 3A is when the cancer is still on the same side of the chest as the tumor, while 3B is when it has spread to the opposite side or near the collarbone. Stage 3 lung cancer is advanced and usually needs a combination of treatments. **Stage IV** is the most serious stage. In this stage the cancer has spread to distant parts of the body, such as the bones, liver, brain, or the other lung. This is called metastatic cancer and it is the hardest to treat.

Small cell lung cancer (SCLC), also a type of lung cancer, is less common but more aggressive. SCLC has a different staging system, instead of four stages, it is separated into the “limited stage” and the “extensive stage”. In the limited stage, the cancer is contained in one lung and nearby lymph nodes. In the extensive stage, the cancer spread throughout the chest or to other organs.

Treatment

Treatment depends on the type and stage of cancer and on the person’s health and wishes. A team may include surgeons, cancer doctors, radiation doctors, nurses, and other staff. They may suggest surgery, radiation therapy, chemotherapy, targeted therapy, immunotherapy, or a mix of these. In many people with early NSCLC, surgery removes the lobe of the lung that holds the tumor. In people with small tumors or weaker lungs, the surgeon may remove a smaller part of the lung. If surgery is too risky, the team may use very focused radiation instead. One method is

called stereotactic body radiotherapy (SBRT). SBRT can sometimes treat small tumors with the aim to cure. Radiation therapy uses high-energy beams to kill cancer cells or slow their growth. It can be used alone when surgery is not possible. It can also be used with chemotherapy when the cancer is too advanced for surgery but still mainly in the chest. Radiation is also very important in SCLC because these cancer cells are very sensitive to both radiation and chemotherapy. Radiation can also ease symptoms. It can reduce bone pain when cancer spreads to bone. It can shrink a tumor that presses on the airway and makes breathing hard.

Chemotherapy uses strong medicines that travel in the blood and reach cancer cells throughout the body. Doctors may give chemotherapy before surgery to shrink a tumor. They may give chemotherapy after surgery to kill any cells that remain. Chemotherapy can also be the main treatment when cancer has already spread. In SCLC, chemotherapy with radiation is often the first treatment. In the last several years, new drugs have changed care for many people with advanced NSCLC. Targeted therapies are pills or infusions that block specific changes in cancer cells. These drugs can work very well when the tumor has a matching gene change. They can shrink tumors and keep the cancer under control for months or years. Immunotherapy is another important type of treatment. Immunotherapy helps the body's own immune system see and attack cancer cells. Many of these drugs block "checkpoints" such as PD-1 or PD-L1. These checkpoints normally act like brakes on the immune system. When the brakes are blocked, immune cells can attack cancer cells more strongly. For many people with advanced NSCLC, doctors now use immunotherapy alone or with chemotherapy as a first treatment.

Care for lung cancer is not only about fighting the tumor. Palliative and supportive care are also very important. Palliative care does not mean that treatment has stopped. It means that the team pays close attention to comfort. Palliative care aims to reduce pain, cough, shortness of

breath, tiredness, and stress. It also supports mood, sleep, nutrition, and family needs. Palliative care can start early, even while a person is getting surgery, chemotherapy, or radiation. Early palliative care can improve daily life. Some research suggests that it may also help people with advanced lung cancer live longer.

Prevention

The chance of developing lung cancer can be lowered by trying to avoid risk factors (factors that increase the chance of lung cancer) such as smoking and doing more protective factors such as regular exercise (factors that decrease the chance of lung cancer). The most common risk factors include smoking, family history, and environmental risk factors. Though lung cancer due to inheritance genes can not be avoided, not smoking and preventing exposure to environmental risk factors may be helpful protective factors. Smoking causes more than half of lung cancer cases, and study shows smokers have 20 times the chance of getting lung cancer than non-smokers. Therefore, by not smoking or quitting smoking, it helps lower the risk of getting lung cancer dramatically. For non-smokers, it is also important to avoid second hand smoke as it exposes the same agents that cause lung cancer in smokers, only in smaller amounts. The other very important factor is radiation, pollution and other environmental factors. Radiation exposure is a very common factor, it includes atomic bomb radiation, radiation therapy, imaging tests, and radon (National Cancer Institution). Other environmental risk factors that should be avoided are carcinogens at work (toxic chemicals that workers were often exposed to) (such as chromium and nickel) and air pollution, which can all cause lung cancer even to non-smokers. Some protective factors include healthy diets and regular physical activity.

Conclusion

In conclusion, the most common causes of lung cancer are smoking, air pollutants, and genetics. A person with lung cancer may experience symptoms such as persistent coughing, chest pain, or shortness of breath. Lung cancer stages are classified into many stages, with each successive stage meaning the cancer is becoming more widespread throughout the body. Treatments are chosen based on the type and stage of lung cancer, including surgery, chemotherapy, and radiation therapy. To prevent developing lung cancer, it is best to avoid smoking/vaping, reduce exposure to air pollution, and maintain an overall healthy lifestyle with good eating and exercise habits.

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Edited Version (in-progress):

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Causes

Lung cancer originally develops when a mutation occurs in a previously normal cell and leads to unregulated cell division, resulting in the production of too many cells. The abnormal multiplication of cells eventually forms a mass called a tumor. There are several causative factors of developing lung cancer. The most common cause of lung cancer, being responsible for about 80% of lung cancer deaths, is smoking. Cigarettes contain over 7,000 chemicals, many of which

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Tests

Once a doctor confirms a person's symptoms are likely lung cancer they first order a chest X-ray. The chest X-ray is a quick and simple way for the doctor to detect possible tumors, however the scan often can miss small tumors. For this reason the doctor will likely order one of

the following tests: LDCT scan, CT scan, or PET-CT scan. An LDCT scan is a special CT scan that uses less radiation than a regular CT, but is still efficient at finding early detections. Though a doctor may use a regular CT scan instead because it gives a clearer and better detailed picture of the lungs and nearby lymph nodes. The doctor may also ask for a PET-CT scan which combines CT photos with images provided by the PET scan. This scan is very good at showing areas where cancer may be active. Occasionally, if there is concern about the lung cancer spreading to the brain or spinal cord, the doctor may order an MRI of the brain.

Although scans give many clues, they cannot prove that a lung mass is definitely cancer. To be sure it's lung cancer, doctors need a biopsy—a process where a small piece of tissue or cells are taken from the body and sent to a lab. A specialist called a pathologist will look at the biopsy sample and decide if it contains cancer and will determine the type of lung cancer as well. The most common biopsy for lung cancer is a bronchoscopy which is when the doctor passes a thin, flexible tube with a light and camera through the mouth or nose and into the airway to check for tumors. If the doctor finds a tumor or mass they will remove tiny pieces of it to send to the pathologist. If the cancer is thought to be in the nodules near the outside of the lung, the healthcare team may use a needle biopsy, where the doctor uses CT images to guide a needle through the chest wall to the spot in the lung. Other times a doctor may call for a surgical biopsy where a small piece of the lung is removed. In a few cases, a sputum cytology can be performed to determine if a person has lung cancer. For this test the patient is asked to cough up a sample of mucus for the pathologist to analyze.

Stages

The staging system that is most often used for NSCLC (non-small cell lung cancer), a type of lung cancer, is the TNM system. There are three factors to it, and the doctors use them to describe how far NSCLC has spread.

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Small cell lung cancer (SCLC), also a type of lung cancer, is less common but more aggressive. SCLC has a different staging system, instead of four stages, it is separated into the “limited stage” and the “extensive stage”. In the limited stage, the cancer is contained in one lung and nearby lymph nodes. In the extensive stage, the cancer spread throughout the chest or to other organs.

Treatment

Treatment depends on the type and stage of cancer and on the person’s health and wishes. A team may include surgeons, cancer doctors, radiation doctors, nurses, and other staff. They may suggest surgery, radiation therapy, chemotherapy, targeted therapy, immunotherapy, or a mix of these. In many people with early NSCLC, surgery removes the lobe of the lung that holds the tumor. In people with small tumors or weaker lungs, the surgeon may remove a smaller part of the lung. If surgery is too risky, the team may use very focused radiation instead. One method is called stereotactic body radiotherapy (SBRT). SBRT can sometimes treat small tumors with the aim to cure. Radiation therapy uses high-energy beams to kill cancer cells or slow their growth. It can be used alone when surgery is not possible. It can also be used with chemotherapy when

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palliative care can improve daily life. Some research suggests that it may also help people with advanced lung cancer live longer.

Prevention

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